

Module 0: 2 Lab Assignment: FAIR Evaluation of Cell Phone Usage Dataset

Curating DATA for A.I. Optimiz-14060ITAI-3375

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Fair Evaluation Of A Public Dataset as Strategic Asset

DataSet selected Global Mobile Phone Addiction Dataset

Area of Focus: Data Science, Consumer Technology, Digital Behavior

Overview: Comprises 700 entries featuring comprehensive metrics on mobile phone usage, which include app usage duration, screen-on duration, battery consumption, data consumption, total number of applications installed, device model, operating system, and classification of user behavior (light/heavy user). This dataset is highly valuable for examining patterns in cell phone usage.

FAIR Principles Scoring (1-5 scale)

- Findable: 4/5

The dataset is readily discoverable on Kaggle, featuring effective search capabilities, relevant tags, and personalized recommendations. It includes a clear title and description but lacks a formal persistent identifier (DOI).

- Accessible: 5/5

The dataset is available for free public download in a clean CSV format, with no login required and an open license.

- Interoperable: 4/5

The dataset is provided in a standard CSV format, ensuring compatibility with most analytical tools such as Python, R, and Excel. The well-structured columns facilitate easy integration.

- Reusable: 4/5

The dataset offers comprehensive variable documentation and is appropriate for machine learning applications. It is licensed for reuse, although it has a limitation due to a small sample size.

Overall FAIR Score: 17/20

Valuation (Cost-Based Approach)

Estimated Value: \$20,000 – \$50,000

This encompasses the expenses associated with gathering authentic user behavior data, as well as the processes of cleaning, labeling, and documentation. The publicly available dataset offers significant educational and prototyping benefits at no cost.

Major Weakness and Suggestions

Major Weakness: The dataset is small, comprising only 700 records, and exhibits limited variability in real-world scenarios. It is not suitable for production use due to its scale and the absence of longitudinal data.

To Enhance Production Readiness for AI Initiatives:

- Expand the dataset using real anonymized mobile logs sourced from carriers or applications.
- Implement versioning, comprehensive metadata, bias audits, and privacy safeguards.
- Incorporate temporal data and connect with additional demographic information

Executive Summary

The dataset on Mobile Device Usage serves as a robust and readily available resource for examining patterns of cell phone usage and user behavior. It achieves a commendable score on the FAIR principles (17 out of 20) and provides significant value for educational purposes and initial AI prototyping endeavors.

Recommendation: This dataset is an excellent foundation for analysis. It is advisable to complement it with larger datasets from real-world scenarios for AI projects intended for production.

Action Items:

1. Download and investigate the CSV file.
2. Develop basic models to forecast user behavior.
3. Leverage insights to gain a deeper understanding of trends in mobile engagement.

Link: <https://www.kaggle.com/code/kennyqq/devices-and-user-behavior>